Problem B - The Perpetual Smuggler's Run

Description

The galaxy is vast and unforgiving. You possess a star map detailing a set of planets connected by a chaotic network of one-way wormholes. Standard wormhole jumps burn through your ship's hyper-fuel, but a few of these spacetime anomalies are saturated with volatile energy—passing through them actually recharges your fuel cells.

The Smuggler's Syndicate has realized that by carefully plotting a course, a pilot could enter a continuous, one-way circuit of planets where the energy gained from volatile wormholes outweighs the fuel spent. A pilot trapped in one of these routes could fly in circles forever, farming infinite fuel.

The Syndicate has further observed that these anomalies are not limited to a single path. If a group of planets is so densely interconnected that every planet in the group can reach every other planet, the presence of even one fuel-generating loop within that group saturates the entire cluster. In such a sector, a pilot can harvest fuel from the anomaly and use the surplus to maintain a perpetual presence across every planet in that interconnected web.

Your ship's nav-computer has exactly 1 second before the Galactic Authorities trace your signal.

Your task: scan the entire star map and output every distinct cluster of planets that forms an infinite-fuel loop.

Input

The first line contains two integers separated by a space: P (the total number of charted planets) and W (the total number of known one-way wormholes). The planets are numbered from 1 to P.

The next W lines detail the wormhole connections. Each line contains three space-separated integers: Origin Planet, Destination Planet, and Fuel Delta.

Output

To successfully program the autopilot before the authorities arrive, your output must follow this strict format:

The first line must contain a single integer L, representing the total number of distinct infinite-fuel loops discovered in the galaxy (may be 0).

The next L lines must list the planets involved in each loop. For each distinct loop, print the IDs of the planets that make up the circuit, separated by spaces, by increasing order.

Constraints

5 <= P <= 60000

5 <= W <= 100000

Example 1

Input:

```
5 5
1 2 10
2 3 5
3 4 -20
4 2 5
5 1 5
```

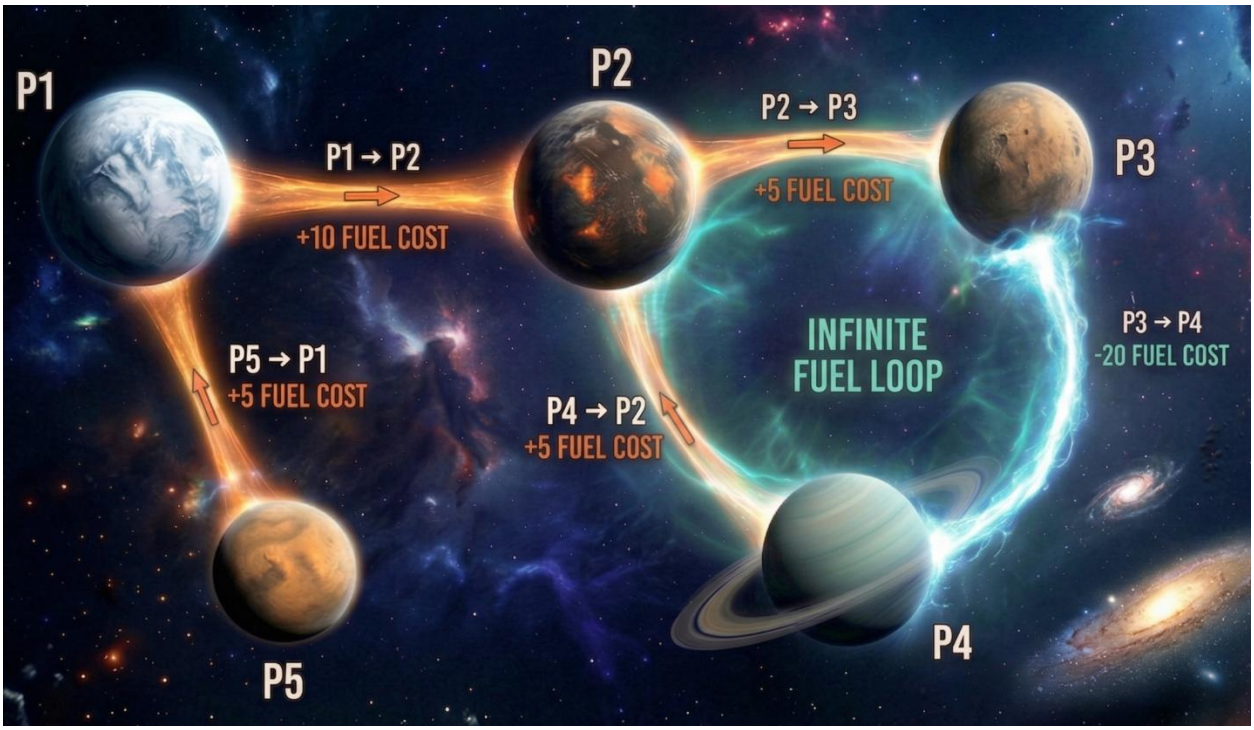
Output:

1

2 3 4

Explanation

In this sector, there are 5 planets and 5 wormholes. Jumping from Planet 1 to 2 costs 10 fuel. However, if a smuggler enters the sequence at Planet 2, they can jump to 3 (costs 5 fuel), jump to 4 (restores 20 fuel), and jump back to 2 (costs 5 fuel). The net result of this circuit is a gain of 10 fuel units. The pilot can loop through planets 2, 3, and 4 forever.



Example 2

Input:

5 6

1 2 10

2 3 5

3 4 -20

4 2 5

5 1 5

3 5 30

Output:

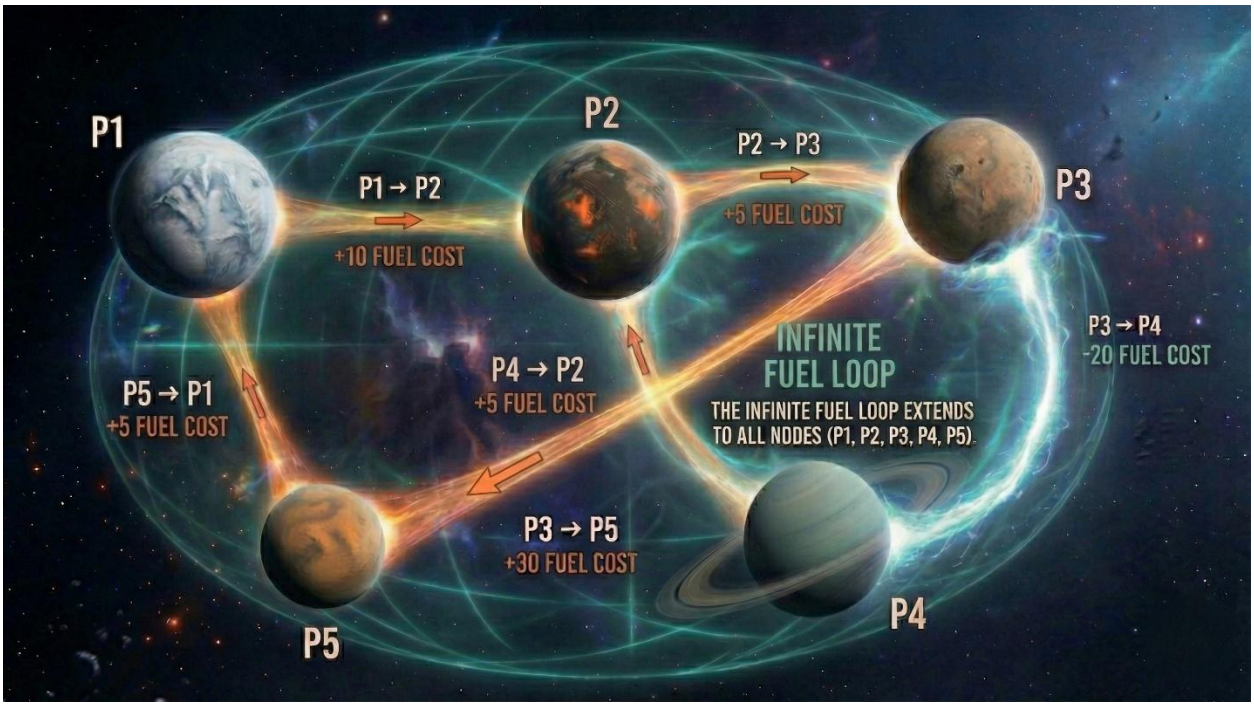
1

1 2 3 4 5

Explanation

In this sector, the core energy-generating loop still exists between Planets 2, 3, and 4 (net gain of 10 fuel). However, notice the addition of the wormhole from Planet 3 to 5 and Planet 5 to 1.

Because Planet 1 leads to Planet 2, and the path from Planet 4 eventually flows back through the network to Planet 1, these five planets now form a single, fully interconnected cluster. Since a pilot can gain infinite fuel within the 2-3-4 cycle and then use that surplus to reach any other planet in this web (and return to the cycle safely), the entire group {1, 2, 3, 4, 5} is identified as one continuous infinite-fuel zone.


P1
P2
P3
P4
P5
P1 → P2
+10 FUEL COST
P2 → P3
+5 FUEL COST
P3 → P4
-20 FUEL COST
P4 → P2
+5 FUEL COST
P5 → P1
+5 FUEL COST
P3 → P5
+30 FUEL COST
INFINITE
FUEL LOOP
THE INFINITE FUEL LOOP EXTENDS
TO ALL NODES (P1, P2, P3, P4, P5).